

Pulse / Loom: A Research Dossier for a Textile-Metaphor P2P Knowledge Fabric

TL;DR

- **Area 1 (code):** Every layer of the proposed "Loom" stack maps onto mature open-source P2P code. The single most important caveat: the canonical Python port, **py-libp2p**, is explicitly "[under development]" and not production-ready, so the highest-fidelity references (rust-libp2p, go-libp2p, Kubo/IPFS) are in Rust/Go and will need translation. Production-grade Python primitives do exist for the DHT (bmuller/kademlia), WebRTC (aiortc), crypto (pynacl/coincurve), and graphs (networkx/rustworkx).
 - **Area 2 (literature):** The metaphor's claims ("self-healing fabric," "voting via Pulses," "free-rider resistance") are each grounded in canonical, citable work — Kademia (2002), Chord (2001), BitTorrent (Cohen 2003), SWIM gossip (2002), CRDTs (Shapiro 2011), Avalanche metastable consensus (2018) — plus hard empirical measurement studies (Adar & Huberman's ~70% free-riders; Saroiu/Gummadi/Gribble; Steiner's KAD crawls).
 - **Area 3 (branding):** "Pulse" is the most dangerous name — **PulseChain (PLS)** is a high-profile, litigated Richard Heart ecosystem. "Loom" is taken by a real Ethereum L2 (LOOM). But "Knit," "Plexus," "Textus," "Silk," "Braid," and "Knot" are effectively free at any meaningful market-cap level — the strongest collision among them is a private (token-less) payments company called Mesh.
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Key Findings

1. **The metaphor is implementable today, mostly by translating Rust/Go.** libp2p is the natural "Mesh + Loom" substrate, Kademia is the natural "Crawler + DHT," IPFS/CIDs supply the "Fibre" (content-addressed hash), CRDTs supply the "Plexus/Textus" convergence math, and SWIM/gossipsub supply the "Pulse." Only the DHT, crypto, WebRTC, and graph layers have battle-tested Python; the orchestration layer (libp2p) does not.
 2. **"Consensus without blockchain" is well-precedented.** Avalanche's metastable, leaderless, subsampled-voting family (Slush→Snowflake→Snowball→Avalanche) is the closest academic analogue to "voting/ingest via Pulses," and CRDTs provide coordination-free convergence ("self-healing fabric") with formal guarantees.
 3. **The empirical record is rich enough for a methodology section.** Decades of measurement studies quantify free-riding, churn, session length, and node counts — exactly the "empirical evidence" a 500-page document needs.
 4. **Branding risk is highly term-dependent.** Rename away from "Pulse" and "Loom" if legal/market distinctiveness matters; "Knit," "Plexus," and "Textus" are open.
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Details

AREA 1 — Technical / Python Implementation

libp2p (the "Mesh" connection layer + "Loom" protocol/validation layer)

- **py-libp2p** — Python. Repo: <https://github.com/libp2p/py-libp2p>. The official org describes it as "The Python implementation of the libp2p networking stack 🐍 [under development]." It supports TCP/WebSockets/QUIC, Noise/TLS security, Mplex multiplexing, and DHT-based peer routing, but is explicitly a work-in-progress aiming for interoperability with Go/Rust/JS. **Maturity: experimental.** Map: this is the **Loom** (protocol rules) + **Mesh** (flat connection layer).
- **rust-libp2p** — Rust, MIT license. Repo: <https://github.com/libp2p/rust-libp2p> (the org page shows ~5,500 stars). The most actively developed reference implementation. **Will need translation to Python.**
- **go-libp2p** — Go. Repo: <https://github.com/libp2p/go-libp2p>. The implementation underpinning IPFS/Kubo. **Will need translation.**
- **jvm-libp2p** — Kotlin/JVM (Apache-2.0), <https://github.com/libp2p> (jvm-libp2p), relevant if a JVM port is ever desired.

Kademlia DHT (the "Crawler" walking the graph + distributed "Fibre" lookup)

- **bmuller/kademlia** — Python (asyncio). Repo: <https://github.com/bmuller/kademlia>. "An asynchronous Python implementation of the Kademlia distributed hash table ... nodes communicate using RPC over UDP ... capable of working behind a NAT." Aims to be a faithful reference implementation of the paper (notes it implements all aspects "except ... republish every 24 hours"). **Maturity: the most usable pure-Python DHT.** Map: the **Crawler/Pulse** routing substrate.
- Canonical references to translate/consult: **go-libp2p-kad-dht** (Go, <https://github.com/libp2p>) and the Kademlia logic inside **rust-libp2p**.

BitTorrent / Mainline DHT (proven large-scale "Crawler" + content distribution)

- **libtorrent** (Rasterbar) is the canonical C++ implementation underpinning most BitTorrent clients; the **BitTorrent BEPs** are the spec corpus (BEP-5 = Mainline DHT, based on Kademlia; BEP-9 = metadata/magnet; PEX). Bram Cohen's economic design is in the 2003 paper (see Area 2). C++ → **needs translation, but the BEPs are the authoritative protocol spec.**

IPFS / content addressing (the "Fibre" = SHA-256 content hash; "Blob" = local store)

- **Kubo** — Go. Repo: <https://github.com/ipfs/kubo>. "Kubo was the first IPFS implementation and is the most widely used one today ... content-addressing (CIDs, DAGs) ... UnixFS ... Bitwap." Map: CIDs are the literal **Fibre** (cryptographic thread/token); Bitwap block exchange is the **Knit** (edge/transfer); the local blockstore is the **Blob** ("ball of wool").

- **multiformats/cid** — spec + libs. Repo: <https://github.com/multiformats/cid>. "A CID is a self-describing content-addressed identifier ... uses cryptographic hashes to achieve content addressing." This is the precise formalization of a **Fibre**.
- For Python: `ipfshttpclient` (talks to a Kubo daemon) rather than a native Python IPFS node.

Gossip / epidemic protocols (the "Pulse" propagation signal)

- **SWIM** is implemented in production by **HashiCorp memberlist/Serf** (Go) — "a decentralized cluster membership tool ... uses a modified SWIM protocol." Repo lineage: github.com/hashicorp/memberlist. Go → **needs translation**, but the algorithm is simple. Map: SWIM's randomized probe + piggybacked dissemination is exactly the **Pulse**.
- **gossipsub** lives in **libp2p pubsub** ([go-libp2p-pubsub](https://github.com/libp2p), <https://github.com/libp2p>). Map: topic-based **Pulse** broadcast.
- Apple's **swift-cluster-membership** is a clean modern SWIM reference (Swift).

CRDTs (the "Plexus/Textus" convergent knowledge graph; "Braid" = merge of streams)

- **Automerge** — Rust core with JS/WASM bindings. "Automerge is a CRDT implementation with a JSON data model ... implemented in Rust, with bindings to JavaScript ... and various other languages." Listed at <https://crdt.tech/implementations>. Rust → **bindings exist; Python via FFI**.
- **Yjs** — JS; the dominant web CRDT (YATA algorithm). <https://crdt.tech/implementations>.
- **GUN** — "a graph CRDT implemented in JavaScript, using WebRTC networking ... public-key-based user authentication and encryption." Directly analogous to a **Plexus** with **Knit** edges.
- Academic reference + implementation catalog: <https://crdt.tech/papers.html> (LWW-register, OR-set, delta-CRDTs).

WebRTC / NAT traversal (the "Mesh" transport across NATs; hole-punching)

- **aiortc** — Python (asyncio). Repo: <https://github.com/aiortc/aiortc>. "A library for Web Real-Time Communication (WebRTC) ... built on top of asyncio ... the aiortc implementation is fairly simple and readable." Supports STUN/TURN/ICE via `RTCIceServer`/`RTCCConfiguration`. **Maturity: usable, the standard Python WebRTC.** Caveat from the issue tracker: it historically lacked ICE-trickle and STUN/TURN failover, so NAT traversal can be brittle.
- **libp2p DCUtR** (Direct Connection Upgrade through Relay) is the canonical hole-punching protocol (in [go/rust-libp2p](https://github.com/libp2p)).

mDNS / Zeroconf + DNS-tunneling fallback

- **python-zeroconf** (pure-Python mDNS/DNS-SD) for LAN peer discovery — the local-mesh **Knot** discovery.

- **iodine / dnscat2** as referenced exotic fallback transports (DNS tunneling) — cite as last-resort transport, not core.

Geospatial hashing (geo-sharding the "Mesh"/"Plexus")

- **Uber H3 (h3-py)** — Python bindings to the H3 hexagonal hierarchical geospatial index. Repo: <https://github.com/uber/h3-py> (core: <https://github.com/uber/h3>, Apache-2.0).
- **Geohash** and **S2 geometry** (google/s2geometry, C++ with Python bindings) are the alternatives.

Graph libraries (representing the "Plexus"/"Web")

- **networkx** — pure Python, the default for the knowledge graph / **Plexus** representation.
- **rustworkx** (formerly retworkx) — Rust-backed (PyO3) graph library from IBM/Qiskit; much faster for large graphs. Repo: <https://github.com/Qiskit/rustworkx>.

Signing / hashing (the cryptographic "Fibre" and "Knit" handshake)

- **pynacl** (libsodium bindings) — ed25519 signatures, the natural peer identity.
- **coincurve** — secp256k1 (Bitcoin/Ethereum-style) signatures.
- **cryptography** (pyca) — general primitives incl. SHA-256 (the literal **Fibre** hash) and Merkle/hash-chain building blocks.

Asyncio concurrency (the engine of the "Crawler" and "Pulse")

- **asyncio** (stdlib), **Trio**, and **anyio** are the three patterns; bmuller/kademlia and aiortc are both asyncio-native, so asyncio is the path of least resistance.

Existing P2P-graph crawlers (prior art for the "Crawler")

- BitTorrent **Mainline DHT crawlers** are the most-studied "crawler that walks a P2P graph" (see Wang & Kangasharju and Steiner in Area 2). Historical Gnutella crawlers underlie the Saroiu/Adar measurement work.

AREA 2 — Academic Literature, Methodology & Empirical Evidence

Foundational DHT papers (the "Crawler"/routing math)

- **Kademlia** — Maymounkov & Mazières, IPTPS 2002. "A peer-to-peer distributed hash table with provable consistency and performance ... using a novel XOR-based metric topology." PDF: <https://www.scs.stanford.edu/~dm/home/papers/kpos.pdf>; Springer: https://link.springer.com/chapter/10.1007/3-540-45748-8_5. 160-bit node IDs; PING/STORE/FIND_NODE/FIND_VALUE RPCs. Cited 3,000+ times.
- **Chord** — Stoica, Morris, Karger, Kaashoek, Balakrishnan, SIGCOMM 2001. PDF: https://pdos.csail.mit.edu/papers/chord:sigcomm01/chord_sigcomm.pdf; DOI:

<https://dl.acm.org/doi/10.1145/383059.383071>. Consistent hashing + finger tables; logarithmic lookup.

- **Pastry** (Rowstron & Druschel 2001), **CAN** (Ratnasamy et al. 2001), **Tapestry** — the other three structured-overlay pillars; cite alongside Chord/Kademlia as the DHT design space.

BitTorrent & incentives (the "Knit" exchange + free-rider resistance)

- **Cohen, "Incentives Build Robustness in BitTorrent," 2003** (Workshop on Economics of P2P Systems / IPTPS'03). PDF: <https://bittorrent.org/bittorrentecon.pdf>. Tit-for-tat, choking/optimistic unchoking, rarest-first. The seminal argument that local incentives produce global robustness — directly supports a "Knit"-level reciprocity model.
- **Measurement: Pouwelse et al., "The BitTorrent P2P File-sharing System: Measurements and Analysis" (IPTPS 2005) and Izal et al. ("Dissecting BitTorrent: Five Months in a Torrent's Lifetime," PAM 2004)** — empirical seeding/leeching and availability data.

Gossip / epidemic (the "Pulse" + "self-healing fabric")

- **Demers et al., "Epidemic Algorithms for Replicated Database Maintenance," PODC 1987** — the origin of anti-entropy/rumor-mongering gossip.
- **SWIM** — Das, Gupta, Motivala, DSN 2002. DOI: 10.1109/DSN.2002.1028914; PDF: https://www.cs.cornell.edu/projects/Quicksilver/public_pdfs/SWIM.pdf. "Scalability ... expected time to first detection of a failure, the false positive rate, and the message load per group member are independent of group size. Time to fully disseminate a failure grows logarithmically with group size. Robustness ... fully partitioned sub-groups can continue to operate, and will automatically merge once connectivity is re-established." This is the literal academic basis for a **self-healing fabric**.
- **Plumtree** (Leitão, Pereira, Rodrigues, SRDS 2007) and **HyParView** (Leitão et al., DSN 2007) — epidemic broadcast trees + membership; the spanning-tree-over-gossip design that a "Pulse" should use.

Eventual consistency & CRDTs (the "Plexus/Textus" convergence)

- **Shapiro, Preguiça, Baquero, Zawirski, "Conflict-free Replicated Data Types," SSS 2011.** Springer: https://link.springer.com/chapter/10.1007/978-3-642-24550-3_29; INRIA TR 7506 (Jan 2011). Formalizes state-based (CvRDT) and op-based (CmRDT) types; "Replicas of any CRDT are guaranteed to converge in a self-stabilising manner, despite any number of failures." A 2018 survey is at <https://arxiv.org/abs/1805.06358>.
- **Amazon Dynamo** — DeCandia et al., SOSP 2007 — vector clocks, quorum, eventual consistency at scale.
- **Lamport, "Time, Clocks, and the Ordering of Events," 1978** — logical clocks; the root of the entire causal-ordering lineage feeding CRDTs.
- **The CALM theorem** (Hellerstein; Ameloot et al.) — monotonic programs need no coordination — the theoretical license for a coordination-free **Plexus**.

Consensus WITHOUT blockchain (grounding "voting/ingest via Pulses")

- **Avalanche / "Snowflake to Avalanche" (Team Rocket, 2018)** and the formal "**Scalable and Probabilistic Leaderless BFT Consensus through Metastability**," **arXiv:1906.08936**. "A family of leaderless Byzantine fault tolerance protocols, built around a metastable mechanism via network subsampling ... quiescent and green ... no node processes more than $O(k)$." The protocol family Slush→Snowflake→Snowball→Avalanche uses repeated random subsampling to steer nodes to consensus — the closest formal analogue to **Pulse-based voting**, and notably it uses a DAG (parents named by each transaction), echoing **Braid**. (arXiv)
- **Hashgraph** (Baird, gossip-about-gossip + virtual voting) — cite as the other leaderless-DAG approach.

Empirical data points (for the methodology / "empirical evidence" section)

- **Free-riding:** Adar & Huberman, "Free Riding on Gnutella," First Monday 5(10), 2000. <https://firstmonday.org/ojs/index.php/fm/article/download/792/701>. From the paper: "By sampling messages on the Gnutella network over a 24-hour period, we established that almost 70% of Gnutella users share no files, and nearly 50% of all responses are returned by the top 1% of sharing hosts" (24-hour weekend trace, Aug 2000; 35,352 hosts sharing 3,304,046 files). They also found ~63% of file-sharing peers got no queries and ~25% of peers provided ~99% of query hits. **Follow-up:** Hughes, Coulson, Walkerdine, "Free Riding on Gnutella Revisited: The Bell Tolls?" (IEEE Distributed Systems Online, 2005) found free-riding rose from ~66% to ~85% by 2005.
- **Heterogeneity / churn / session length:** Saroiu, Gummadi, Gribble, "A Measurement Study of Peer-to-Peer File Sharing Systems," MMCN 2002 (DOI 10.1117/12.449977; PDF <https://people.mpi-sws.org/~gummadi/papers/mmcn.pdf>) — characterized bottleneck bandwidths, latencies, connect/disconnect frequency, and host uptime for Napster and Gnutella; the canonical "peers are highly heterogeneous and many lie about capacity" result. Companion: Gummadi et al., "Measurement, Modeling, and Analysis of a Peer-to-Peer File-Sharing Workload," SOSP 2003 (DOI 10.1145/945445.945475).
- **DHT-scale measurement:** Steiner, En-Najjary, Biersack, "Long Term Study of Peer Behavior in the KAD DHT" (and "A Global View of KAD," IMC 2007, DOI 10.1145/1298306.1298323) — crawled KAD every 5 minutes for ~6 months; found ~1.5M+ simultaneous eMule/KAD users and **two classes of peers** (long-lived weeks-long peers vs. short-lived peers that leave within days). PDF: <https://www.moritzsteiner.de/papers/KADsteiner.pdf>.
- **Mainline DHT scale:** Wang & Kangasharju, "Measuring Large-Scale Distributed Systems: Case of BitTorrent Mainline DHT" (IEEE P2P 2013, Trento). Per the authors' project site (cs.helsinki.fi/u/jakangas/MLDHT): "the number varies between 15 and 27 million per day with a very clear daily churn pattern," with European users dominating.
- **Zipf popularity:** file-popularity in P2P systems follows Zipf/power-law distributions (Sripanidkulchai, "The Popularity of Gnutella Queries..."; Saroiu/Gummadi workload study) — cite for content-availability modeling.

Modern decentralized data systems (comparables for the "Web/Plexus")

- **IPFS** — Benet, "IPFS - Content Addressed, Versioned, P2P File System," arXiv:1407.3561 (2014). Created by Juan Benet / Protocol Labs.
 - **Secure Scuttlebutt** (Tarr et al., "Secure Scuttlebutt: An Identity-Centric Protocol for Subjective and Decentralized Applications," ICN 2019) — append-only logs + gossip, a strong social-graph analogue.
 - **Hypercore/Dat** (Ogden et al., Dat whitepaper) — signed append-only logs.
 - **OrbitDB** — CRDT databases over IPFS pubsub — almost a direct prototype of a "Plexus over Pulse."
 - **Holochain** and **OriginTrail DKG** — agent-centric / decentralized knowledge-graph designs worth contrasting.
 - Useful survey: "IPFS and Friends: A Qualitative Comparison of Next Generation Peer-to-Peer Data Networks," arXiv:2102.12737.
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AREA 3 — Comparative Crypto Terminology & Branding Risk

A subagent verified rankings across CoinMarketCap/CoinGecko/CoinPaprika. The headline: only "**Pulse**" and "**Loom**" are occupied by significant projects; the textile-core terms are essentially free.

"Pulse" — HEAVILY OCCUPIED (highest branding risk)

- **PulseChain (PLS)** — a Layer-1 Ethereum hard fork by **Richard Heart** (founder of HEX), launched May 13, 2023; it copied Ethereum's entire state ("the largest airdrop in history"). Per CoinMarketCap Academy, PulseChain's sacrifice phase "ended on August 22, 2021, and raised over \$700 million for medical research and other causes"; PLS has "a fixed supply of four trillion tokens." The ecosystem includes **PulseX (PLSX)** and **HEX**. CoinMarketCap page: <https://coinmarketcap.com/currencies/pulsechain/>.
- **Litigation context (material to any filing):** The SEC charged Richard Heart and Hex/PulseChain/PulseX on July 31, 2023 (Press Release 2023-143) with offerings that "raised more than \$1 billion in crypto assets" and "misappropriating at least \$12 million of offering proceeds to purchase luxury goods including sports cars, watches, and a 555-carat black diamond known as The Enigma"; the case was dismissed Feb 28, 2025 by U.S. District Judge Carol Bagley Amon on jurisdictional grounds.
- **Note a data conflict to flag:** CoinMarketCap currently shows PLS at rank #5,718 with market cap "not available," while CoinPaprika lists it at ~\$87M market cap, rank ~#267 — and PulseChain's own community sites allege CMC/CoinGecko "manually suppress or exclude PulseChain ecosystem tokens." Regardless of the exact rank, this is a large, high-profile, recently-litigated ecosystem. **Verdict: avoid "Pulse" as the brand name; collision and reputational-association risk are both high.**

"Loom" — OCCUPIED (real L2)

- **Loom Network (LOOM)** — an ERC-20 token for an Ethereum L2 / DPoS sidechain platform ("DAppChains," later PlasmaChain/Basechain), launched 2017, focused on games/social then enterprise. LOOM's all-time high was \$0.7745 on May 4, 2018 (CoinMarketCap, currencies/loom-network); it is now ~\$6.49M market cap at rank #1,126, with a circulating supply of 1,242,920,898 LOOM. Pages: <https://www.coinbase.com/price/loom-network>. Notably its own marketing leans on the textile metaphor ("old-school analog looms used to weave carpets"). **Verdict: "Loom" is taken by a still-listed project in an adjacent (blockchain-infra) space — moderate collision risk.**

"Fiber/Fibre" — PARTIALLY OCCUPIED (concept, not a big token)

- **Nervos Fiber Network** — a Lightning-style payment/swap layer on Nervos CKB; Rust implementation at <https://github.com/nervosnetwork/fiber> ("A scalable, privacy-by-default payment & swap network"), testnet through v0.6.0 with mainnet targeted early 2026. It's a *network/feature* of CKB, not a separately ranked "Fiber" token. Historical "Fibre" coin and FIBOS exist but are negligible. **Verdict: "Fiber" is associated with Nervos's payment layer but not occupied by a large standalone token; moderate risk.**

"Braid" — LIGHTLY OCCUPIED (technical concept, no token)

- **Braidpool** — a decentralized Bitcoin mining-pool proposal using a **merge-mined DAG of "beads" with Nakamoto-like consensus** (successor to P2Pool). Repo: <https://github.com/braidpool/braidpool>; spec: [.../docs/braidpool_spec.md](https://github.com/braidpool/braidpool/blob/master/docs/braidpool_spec.md). "Braidpool is a Directed Acyclic Graph (DAG) of 'beads' rather than a blockchain of 'blocks' ... each bead can have multiple parents, and the graph can have diamonds or other higher-order graph structures." There is "no altcoin, ICO etc." **Verdict: "Braid" is used as a DAG-consensus term but has no token — low collision, and conceptually it directly supports the project's "Braid = interweaving of streams" metaphor.** (GitHub) (Stacker News)

"Weave" — OCCUPIED by Arweave

- **Arweave (AR)** uses "blockweave" and "permaweb." "Each block in the blockweave is linked to two previous blocks: the immediately preceding block and a randomly selected ... 'recall block'." Yellow paper: <https://www.arweave.org/yellow-paper.pdf>. AR is a top-tier storage token. **Verdict: avoid "Weave"; strongly associated with Arweave.** (ZeroCap)

Terms that are effectively FREE (low branding risk)

- **"Knit"** — only **Knit Finance (KFT)**, a micro-cap cross-chain wrapped-asset protocol (<https://coinmarketcap.com/currencies/knit-finance/>), far outside the top-600. "Knit Finance is a protocol for creating cross-chain wrappers." **Free at scale.** (CoinGecko)
- **"Knot"** — only dead/tiny tokens: **Karmaverse (KNOT)**, a GameFi token that "stopped trading"; and **Knot Diffie-Hellman (KNOT)**, a post-quantum project at CMC rank

~#7,563. Blockswap uses "KNOT" as a *staking-unit concept*, not a top-600 ticker. **Free.**

[CoinGecko](#)

[CoinMarketCap](#)

- **"Mesh"** — fragmented across small tokens (**Mesh Protocol/MESH** on Solana ~#4,061; **Meshswap**; **SmartMesh/SMT**; **MeshChain**). The real brand-collision is a **private, token-less payments company, Mesh** (founded 2020, San Francisco; co-founders Bam Azizi and Adam Israel). Per PR Newswire (Aug 14, 2025): "additional investments from PayPal Ventures, Coinbase Ventures, Uphold, Mirana Ventures... bringing total capital funding to more than \$130 million"; its \$82M Series B (March 11, 2025) was led by Paradigm and settled largely in PayPal USD (PYUSD). **No ranked top-600 MESH token, but the well-funded private company is a real brand-collision watch item.**
 - **"Plexus"** — only **PLEXUS (PLX)**, a micro-cap Arbitrum cross-chain aggregator (<https://coinmarketcap.com/currencies/plexus/>). "PLEXUS is a ... cross-chain aggregator ... just like PLEXUS inside the human body." **Free at scale.** [CoinMarketCap](#)
 - **"Spin"** — **Spin.fi** (NEAR order-book DEX) has *no live token yet*; **Spintop (SPIN)** is a dead micro-cap (~#12,085). **Free.**
 - **"Silk"** — **Silk (SILK)** is a niche privacy stablecoin by **Shade Protocol** on Secret Network (governance token SHD ~#3,842). The historical **"Silk Road" darknet market is NOT a token** — distinguish clearly. **Free at scale.**
 - **"Saito" (SAITO)** — a recognizable-name but small (~#1,700, ~\$4M cap) Layer-1 P2P/Web3 project that pays routing nodes rather than miners (<https://coinmarketcap.com/currencies/saito/>; <https://github.com/SaitoTech/saito>). Relevant as *prior art* for "pay nodes for relaying" (a possible **Silk/Pulse** incentive), not as a name collision. [CoinMarketCap](#)
 - **"Textus"** — **no cryptocurrency uses it at all.** Closest is Obyte's "textcoins" feature (not a token). **Completely free.**
 - **"Silk"/"Crawler"/"Node"** as generic words are unregistrable but unoccupied as distinctive marks.
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Recommendations

Staged build plan (Area 1):

1. **Prototype phase — stay in Python.** Build the **Knot/Mesh** layer on **bmuller/kademlia** (DHT/Crawler) + **aiortc** (NAT-traversing transport) + **pynacl** (ed25519 Fibre identity) + **networkx** (Plexus). This gets a walking Crawler and a Knit graph running with zero translation. Benchmark threshold to advance: stable 100-node testnet with <5% message loss under churn.
2. **Scale phase — translate selectively.** Port **SWIM** (from HashiCorp memberlist) for the **Pulse**, and swap **networkx**→**rustworkx** when graph ops exceed ~10⁵ nodes. Adopt **Automerge** (via Rust FFI) for the **Plexus/Textus** CRDT once multi-writer convergence is needed.

3. **Production phase — adopt libp2p.** If/when py-libp2p matures (track its releases) or via a Rust-core binding, migrate the **Mesh/Loom** layer to libp2p for interop, gossipsub **Pulse**, and DCUtR hole-punching. Until then, treat rust-libp2p/go-libp2p as the spec to translate, not a dependency.
4. **Consensus:** implement **Pulse voting** as an Avalanche-style subsampled metastable vote (cite arXiv:1906.08936), with CRDT merge as the fallback "self-healing" layer for non-conflicting ingests.

Branding decision (Area 3):

- **Do not ship as "Pulse" or "Loom"** if you need a clean trademark and want to avoid association with the litigated PulseChain/Richard Heart ecosystem or the existing LOOM token. Avoid "Weave" (Arweave).
- **Safest distinctive names: "Knit," "Plexus," "Textus," or a coined compound (e.g., "KnitNet," "Plexus Fabric").** "Textus" is the single cleanest option (zero crypto usage). Keep "Braid," "Silk," "Fibre," and "Crawler" as internal architecture terms (low risk) rather than the top-line brand.
- Before finalizing, run a formal trademark search (USPTO/EUIPO) and a domain/handle sweep — market-cap-free does not mean trademark-free.

Thresholds that change the recommendation:

- If py-libp2p reaches a tagged stable (1.0-class) release, promote it from "translate later" to "build on now."
- If "Pulse"/"Loom" associations are deemed irrelevant for a non-token, non-financial knowledge tool, the naming constraint relaxes — but the existing LOOM/PLS search-result noise still hurts discoverability.

Caveats

- **py-libp2p is not production-ready** ("[under development]"); any plan that assumes a native Python libp2p today is premature. The high-fidelity references are Rust/Go.
- **aiortc** has historically had NAT-traversal gaps (no ICE-trickle, no automatic STUN/TURN failover per its issue tracker); budget engineering for robust hole-punching.
- **PulseChain ranking data conflicts** between sources (CMC "rank #5,718 / cap unavailable" vs. CoinPaprika "~#267 / ~\$87M"), and PulseChain advocates allege deliberate suppression by aggregators — treat any single rank figure as approximate and date-stamped. The SEC case was dismissed on jurisdictional grounds in Feb 2025, not on the merits.
- Crypto rankings are volatile and date-stamped; many cited small tokens (KNOT, SPIN, MESH variants) show near-zero liquidity or have "stopped trading," so deep rankings (>#5,000) are effectively meaningless.

- Several measurement statistics are era-specific (Gnutella 2000–2005, KAD 2007); they establish *patterns* (free-riding, churn classes, Zipf popularity) rather than current absolute numbers.
- A few secondary descriptions were drawn from explainer/news sites rather than primary papers; the seminal claims here are anchored to primary PDFs (Stanford, MIT PDOS, bittorrent.org, Cornell, arXiv, Springer) wherever possible.
- Two technical sub-areas (gossipsub spec deep-dive, H3/rustworkx hands-on benchmarks) were identified but not exhaustively sourced before the search budget closed; the canonical repos are linked and should be the next reading.